

Invoice Analytics Project

End-to-End Data Analytics Project

Submitted By

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Tools Used

Microsoft Excel

MySQL

Python (Pandas, Matplotlib)

Power BI (DAX)

Project Duration

June 2026 – July 2026

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Analysis 1: Project Overview

The **Invoice Analytics Project** is an end-to-end data analytics solution developed to analyze invoice processing data using Excel, MySQL, Python, and Power BI. The project demonstrates the complete analytics workflow, starting from data preparation and database design to business analysis and interactive dashboard creation.

The project uses two datasets:

- **Header_Data:** containing invoice-level information
- **Line_Items_Data:** containing detailed line-item information

The data was first cleaned and analyzed in Microsoft Excel, then stored in a MySQL relational database. SQL queries were used to validate and analyze the data. Python was used for data cleaning, transformation, and visualization. Finally, Power BI was used to build interactive dashboards for business reporting.

This project demonstrates practical skills in data modeling, SQL querying, Python programming, data visualization, and business intelligence.

Analysis 2: Objectives

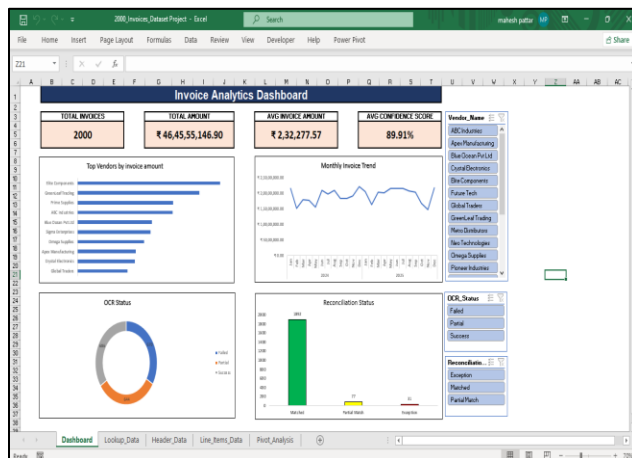
The main objectives of this project are:

- To design a relational database for invoice analytics.
- To analyze invoice and line-item data using SQL.
- To perform data cleaning and exploratory data analysis using Python.
- To create interactive dashboards using Power BI.
- To generate business insights from invoice processing data.
- To demonstrate an end-to-end data analytics workflow.

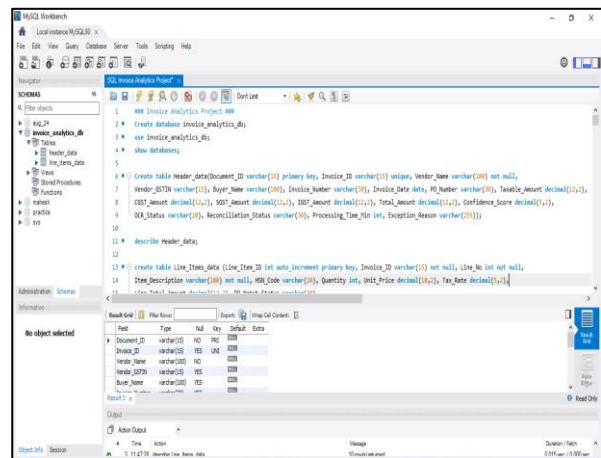
Analysis 3: Tools and Technologies

Tools	Purpose
Microsoft Excel	Data Cleaning & Dashboard
My SQL	Database Design & SQL Queries
Python (Pandas, Matplotlib)	Data Analysis & Visualization, Data Manipulation, Charts
Power BI (DAX)	Dashboard & Reporting, Business Calculations

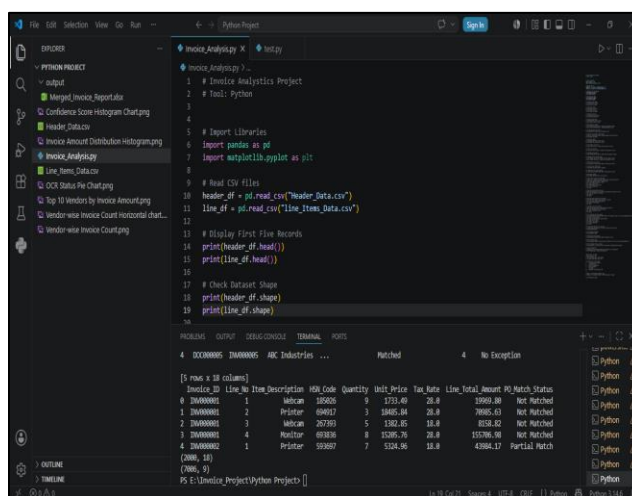
Excel Dashboard



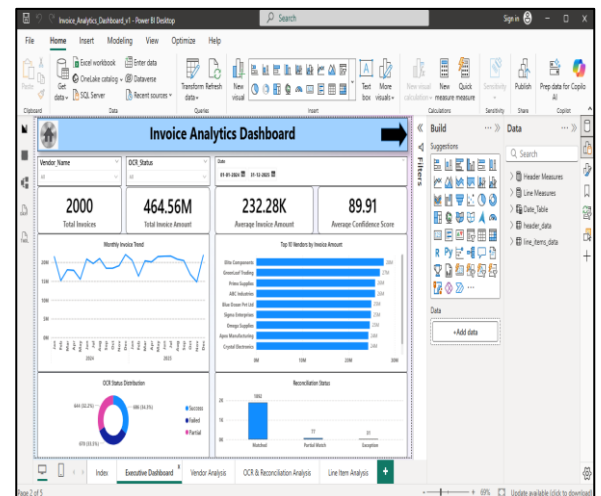
MySQL Database and Tables



Python VS Code



Power BI Dashboard



Analysis 4: Dataset Description

4.1: Dataset Overview

The Invoice Analytics Project uses two datasets that represent invoice header information and invoice line-item details. The datasets were separated to maintain database normalization and establish a one-to-many relationship between invoices and their line items.

4.2: Header Data

The **Header_Data** table stores invoice-level information. Each record represents one invoice and contains details such as vendor information, invoice amount, invoice date, OCR status, confidence score, and reconciliation status.

Figure 4.1: Header_Data Sample

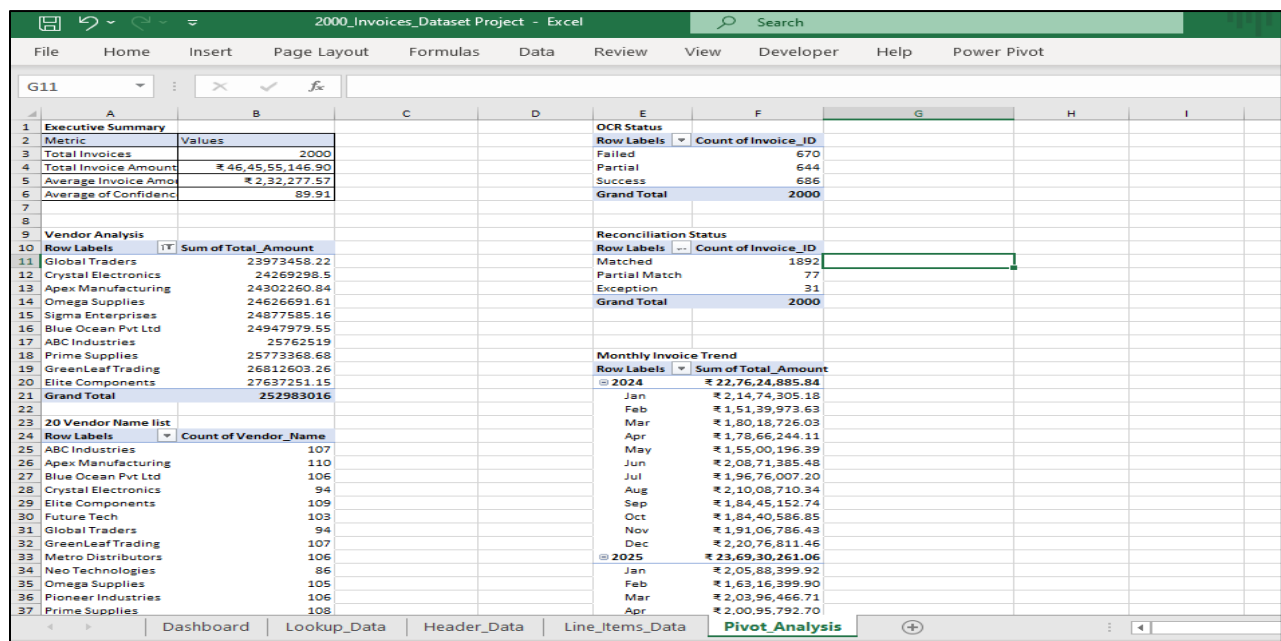
	D	E	F	G	H	I	J	K	L
1	Vendor_GSTIN	Buyer_Name	Invoice_Number	Invoice_Date	PO_Number	Taxable_Amount	CGST_Amount	SGST_Amount	IGST_Amount
2	19ABCDE1679F1Z8	Cognizant	INVNO-100001	07-03-2025	PO-3352480	199619.26	0.00	0.00	552
3	11ABCDE2246F1Z6	Infosys	INVNO-100002	09-06-2025	PO-7734282	167762.14	15098.59	15098.60	
4	14ABCDE3249F1Z5	Infosys	INVNO-100003	24-01-2025	PO-1126762	34186.68	3656.31	3656.31	
5	26ABCDE1853F1Z4	HCL	INVNO-100004	06-05-2025	PO-1820727	184647.84	0.00	0.00	227
6	11ABCDE5094F1Z9	Infosys	INVNO-100005	16-12-2024	PO-3110092	153505.50	0.00	0.00	139
7	35ABCDE2501F1Z7	TCS	INVNO-100006	29-10-2024	PO-3603781	268645.86	26870.24	26870.24	
8	15ABCDE5201F1Z2	Capgemini	INVNO-100007	21-05-2025	PO-6417321	284599.00	0.00	0.00	273
9	21ABCDE2311F1Z7	IBM India	INVNO-100008	12-08-2024	PO-3763982	126175.41	0.00	0.00	353
10	25ABCDE3562F1Z5	Infosys	INVNO-100009	15-09-2025	PO-2680498	349840.11	24196.01	24196.00	

4.3: Line_Items_Data

The **Line_Items_Data** table stores detailed information about each invoice line item. Each invoice may contain multiple line items.

Figure 4.2: Line_Items_Data Sample

	A	B	C	D	E	F	G	H	I	J
1	Invoice_ID	Line_No	Item_Description	HSN_Code	Quantity	Unit_Price	Tax_Rate	Line_Total_Amount	PO_Match_Status	
2	INV000001	1	Webcam	185026	9	1733.49	28.00	19969.80	Not Matched	
3	INV000001	2	Printer	694917	3	18485.84	28.00	70985.63	Not Matched	
4	INV000001	3	Webcam	267393	5	1382.85	18.00	8158.82	Not Matched	
5	INV000001	4	Monitor	693836	8	15205.76	28.00	155706.98	Not Matched	
6	INV000002	1	Printer	593697	7	5324.96	18.00	43984.17	Partial Match	
7	INV000002	2	Tablet	511121	7	18641.06	18.00	153975.16	Not Matched	
8	INV000003	1	Mouse	576979	8	2171.71	28.00	22238.31	Matched	
9	INV000003	2	Mouse	935060	2	3586.88	18.00	8465.04	Matched	
10	INV000003	3	Webcam	554313	2	4819.62	12.00	10795.95	Matched	



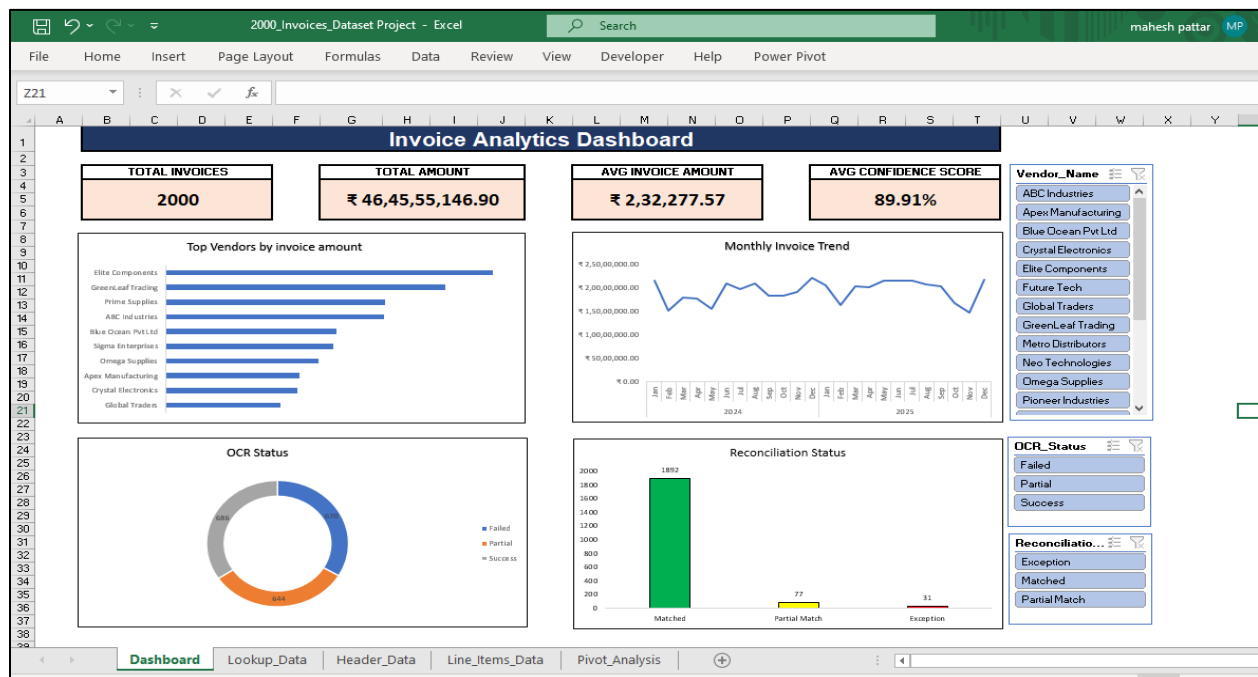
5.3: Excel Dashboard

An interactive dashboard was developed using Pivot Charts, KPI cards, slicers, and conditional formatting to provide business insights.

Dashboard features:

- KPI Cards
- Vendor Analysis
- Monthly Trend
- OCR Analysis
- Top Vendors
- Interactive Slicers

Figure 5.1: Excel Dashboard Sample



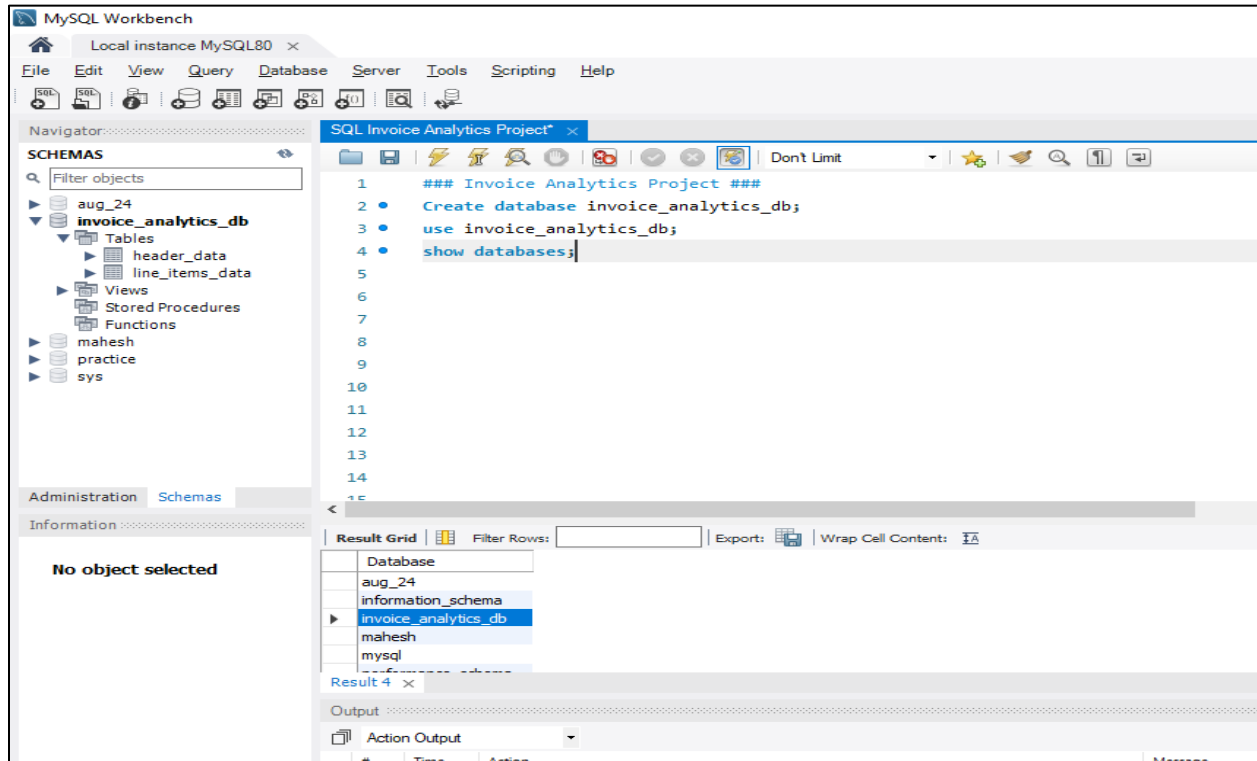
Analysis 6: SQL Database Design & Implementation

6.1: Database Creation

A relational database named `invoice_analytics_db` was created in MySQL to store invoice data. The database consists of two related tables: `Header_Data` and `Line_Items_Data`.

The database design follows normalization principles to reduce redundancy and maintain data integrity.

Figure 6.1: Database Creation Sample



6.2: Table Creation

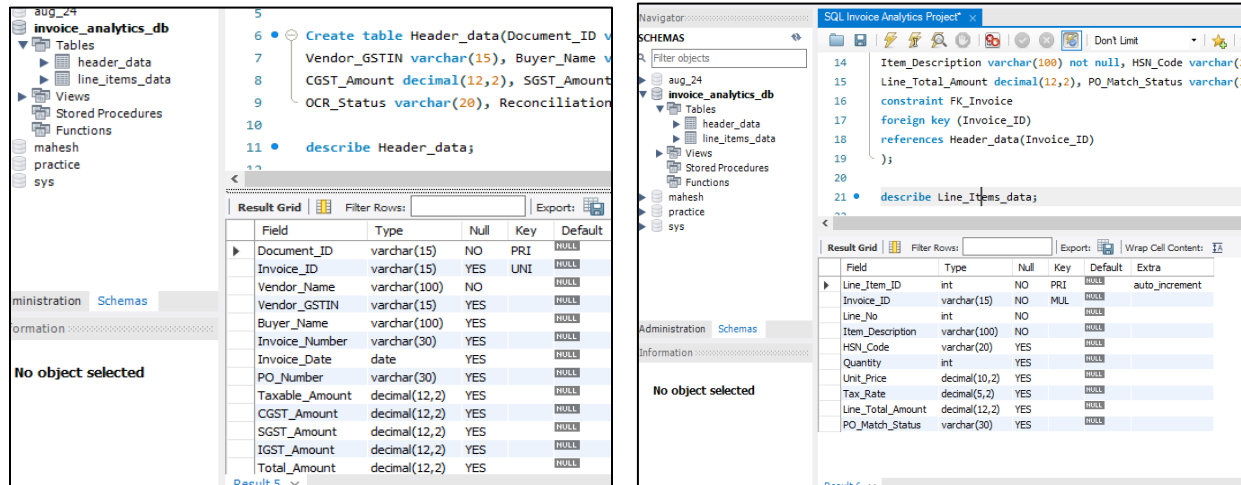
Two tables were created to organize invoice data efficiently.

Header_Data: Stores invoice-level information.

Line_Items_Data: Stores detailed line-item information.

Primary Key and Foreign Key constraints were implemented to establish relationships between the tables.

Figure 6.2: Table Creation Sample



6.3: Data Import

The cleaned datasets were exported as CSV files and imported into MySQL using the Table Data Import Wizard.

Table	Records
Header_Data	2000
Line_Items_Data	7006

6.4: SQL Queries

More than **50 SQL queries** were developed to analyze invoice data and generate business insights.

The queries covered:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- AGGREGATE FUNCTION
- CASE
- JOINS
- Subqueries
- Views
- CTEs
- Window Functions

Figure 6.3: Queries sample

```
SQL Invoice Analytics Project x
80
81  ## Find vendors whose names end with Ltd.
82 •  select * from header_data where vendor_name like "%Ltd";
83
84  ## Display invoices from highest amount to lowest.
85 •  select invoice_id, vendor_name, total_amount from header_data order by total_amount desc;
86
87  ## Display invoices from oldest to newest.
88 •  select invoice_id, invoice_date, total_amount from header_data order by invoice_date;
89
90  ## Display the top 10 invoices by amount.
91 •  select invoice_id, vendor_name, total_amount from header_data order by total_amount desc limit 10;
92
93  ##### ----- Aggregate Functions and GROUP BY ----- #####
94 •  use invoice_analytics_db;
95  ## Vendor-wise Invoice Count
```

6.5: SQL Business Analysis

Queries Included:

- Total number of invoices.
- Vendor-wise invoice amount.
- Vendor-wise invoice count.
- OCR status summary.
- Reconciliation status summary.
- Top vendors by invoice amount.
- High-value invoices.

Figure 6.4: Queries Sample

```
90  ## Display the top 10 invoices by amount.
91 •  select invoice_id, vendor_name, total_amount from header_data order by total_amount desc limit 10;
92
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	invoice_id	vendor_name	total_amount			
▶	INV001846	GreenLeaf Trading	758367.54			
	INV000245	Omega Supplies	706535.96			
	INV001868	GreenLeaf Trading	661023.83			
	INV001077	Global Traders	660951.02			
	INV000570	Tech Solutions	660485.11			
	INV001936	Neo Technologies	644379.47			
	INV000569	Blue Ocean Pvt Ltd	628262.96			
	INV000254	Neo Technologies	628250.28			
	INV001051	Tech Solutions	616410.28			

header_data 7 x

Analysis 7: Python Data Analysis

7.1: Python Introduction

Python was used to perform data loading, cleaning, validation, business analysis, and data visualization. The project used the **Pandas** library for data manipulation and **Matplotlib** for visualization.

7.2: Libraries Used

Library	Purpose
Pandas	Data Loading & Analysis
Matplotlib	Data Visualization
PyMySQL	Connect Python to MySQL
SQLAlchemy	Database Connection

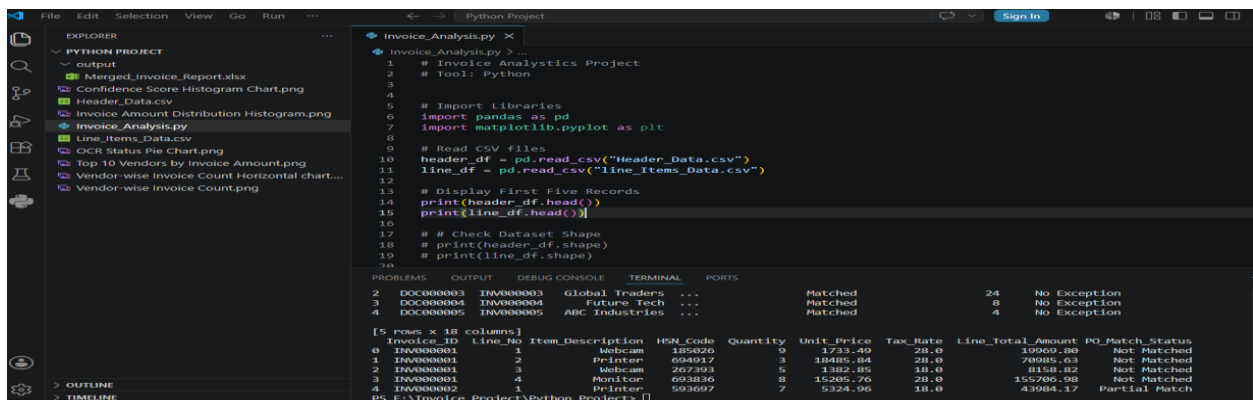
7.3: Data Loading

The invoice datasets were loaded into Python using Pandas. Data was read from CSV files and later connected directly to the MySQL database for analysis.

Activities performed:

- Read Header_Data.csv
- Read Line_Items_Data.csv
- Connected Python to MySQL
- Loaded data into Data Frames

Figure 7.1: Read Header_Data sample



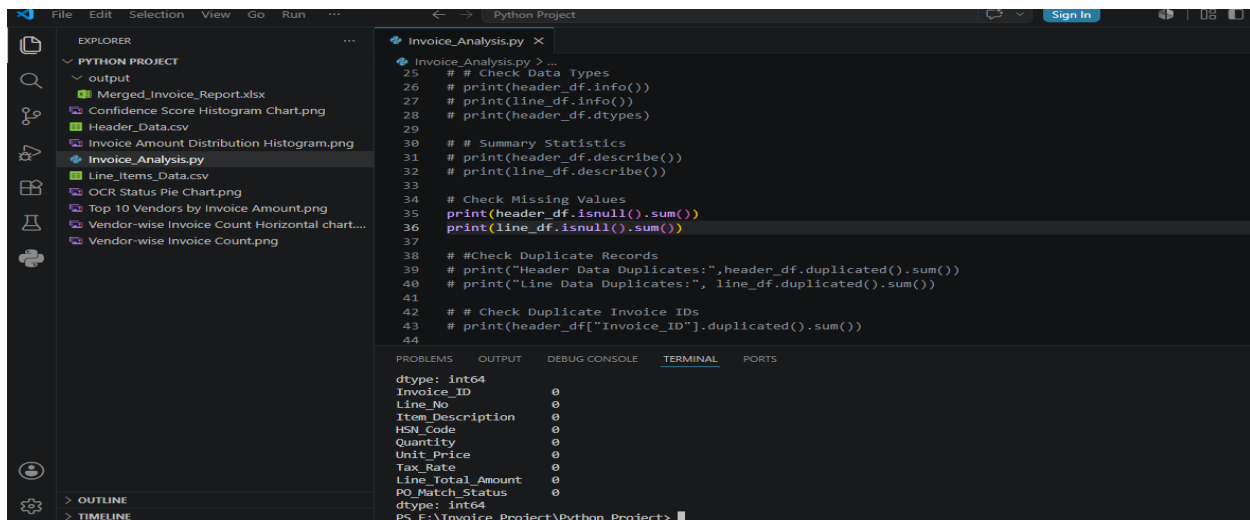
7.4: Data Cleaning

Several validation checks were performed before analysis.

Activities:

- Checked missing values
- Checked duplicate records
- Verified data types
- Converted Invoice_Date to Date format
- Verified date range

Figure 7.2: Checked Missing Values sample



The screenshot shows a Python IDE with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'PYTHON PROJECT' with various files including 'Merged_Invoice_Report.xlsx', 'Confidence Score Histogram Chart.png', 'Header_Data.csv', 'Invoice_Analysis.py', 'Line_Items_Data.csv', 'OCR Status Pie Chart.png', 'Top 10 Vendors by Invoice Amount.png', 'Vendor-wise Invoice Count Horizontal chart...', and 'Vendor-wise Invoice Count.png'. The code editor displays the 'Invoice_Analysis.py' file with the following code:

```
25 # Check Data Types
26 # print(header_df.info())
27 # print(line_df.info())
28 # print(header_df.dtypes)
29
30 # Summary Statistics
31 # print(header_df.describe())
32 # print(line_df.describe())
33
34 # Check Missing Values
35 print(header_df.isnull().sum())
36 print(line_df.isnull().sum())
37
38 # Check Duplicate Records
39 # print("Header Data Duplicates:", header_df.duplicated().sum())
40 # print("Line Data Duplicates:", line_df.duplicated().sum())
41
42 # Check Duplicate Invoice IDs
43 # print(header_df["Invoice_ID"].duplicated().sum())
44
```

The terminal output shows the following results:

```
dtype: int64
Invoice_ID      0
Line_No         0
Item_Description 0
HSN Code        0
Quantity        0
Unit_Price      0
Tax_Rate        0
Line_Total_Amount 0
PO_Match_Status 0
dtype: int64
PS E:\Invoice Project\Python Project>
```

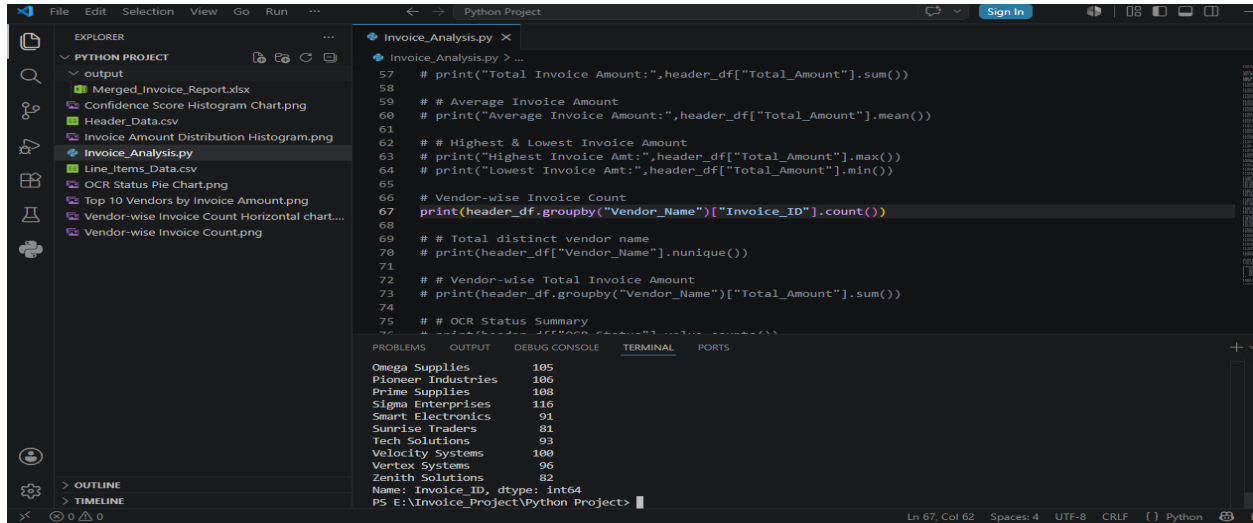
7.5: Data Analysis

Business analysis was performed using Pandas.

Analysis included:

- Total invoices
- Total invoice amount
- Average invoice amount
- Vendor-wise analysis
- OCR status analysis
- Confidence score analysis

Figure 7.3: Vendor-wise Invoice Count sample



```
57 # print("Total Invoice Amount:",header_df["Total_Amount"].sum())
58
59 # # Average Invoice Amount
60 # print("Average Invoice Amount:",header_df["Total_Amount"].mean())
61
62 # # Highest & Lowest Invoice Amount
63 # print("Highest Invoice Amt:",header_df["Total_Amount"].max())
64 # print("Lowest Invoice Amt:",header_df["Total_Amount"].min())
65
66 # Vendor-wise Invoice Count
67 print(header_df.groupby("Vendor_Name")["Invoice_ID"].count())
68
69 # # Total distinct vendor name
70 # print(header_df["Vendor_Name"].nunique())
71
72 # # Vendor-wise Total Invoice Amount
73 # print(header_df.groupby("Vendor_Name")["Total_Amount"].sum())
74
75 # # OCR Status Summary
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Vendor	Count
Omega Supplies	105
Pioneer Industries	106
Prime Supplies	108
Sigma Enterprises	116
Smart Electronics	91
Sunrise Traders	81
Tech Solutions	93
Velocity Systems	100
Vertex Systems	96
Zenith Solutions	82

Name: Invoice_ID, dtype: int64
PS E:\Invoice_Project\Python Project>

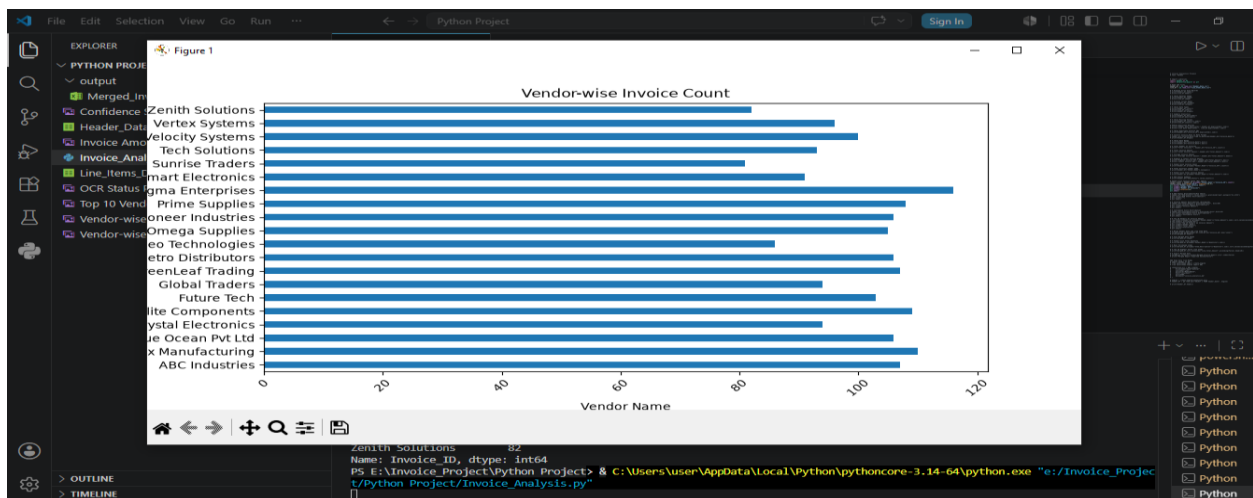
7.6: Data Visualization

Matplotlib was used to create visual representations of the invoice data.

Charts created:

- Vendor-wise Invoice Count
- OCR Status Distribution
- Invoice Amount Distribution
- Confidence Score Distribution
- Top 10 Vendors

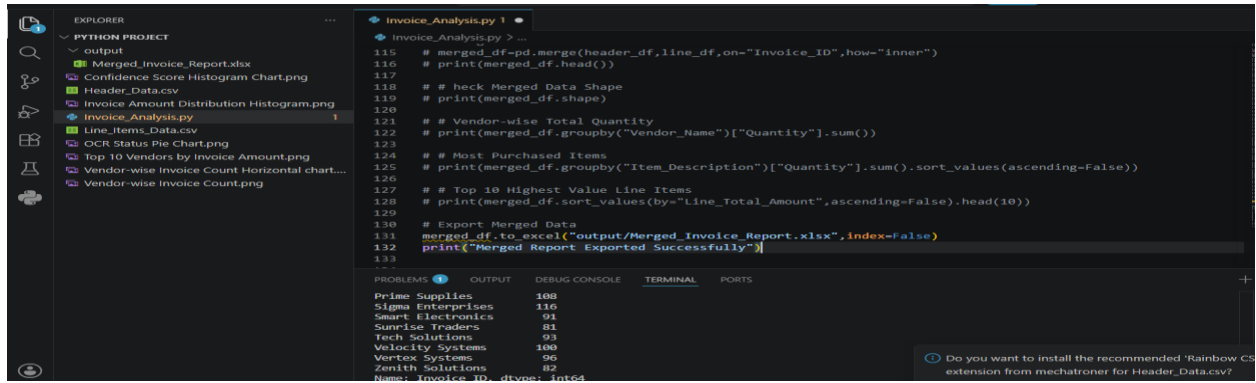
Figure 7.4: Vendor-wise Invoice Count bar chart sample



7.7: Data Export

The merged dataset and analysis results were exported to Excel for reporting and further analysis.

Figure 7.5: Successful exported merged invoice report Sample



Analysis 8: Power BI Dashboard

8.1: Power BI Introduction

Power BI was used to develop an interactive multi-page dashboard for invoice analytics. DAX measures, relationships, slicers, and visualizations were used to provide business insights.

8.2: Data Modeling

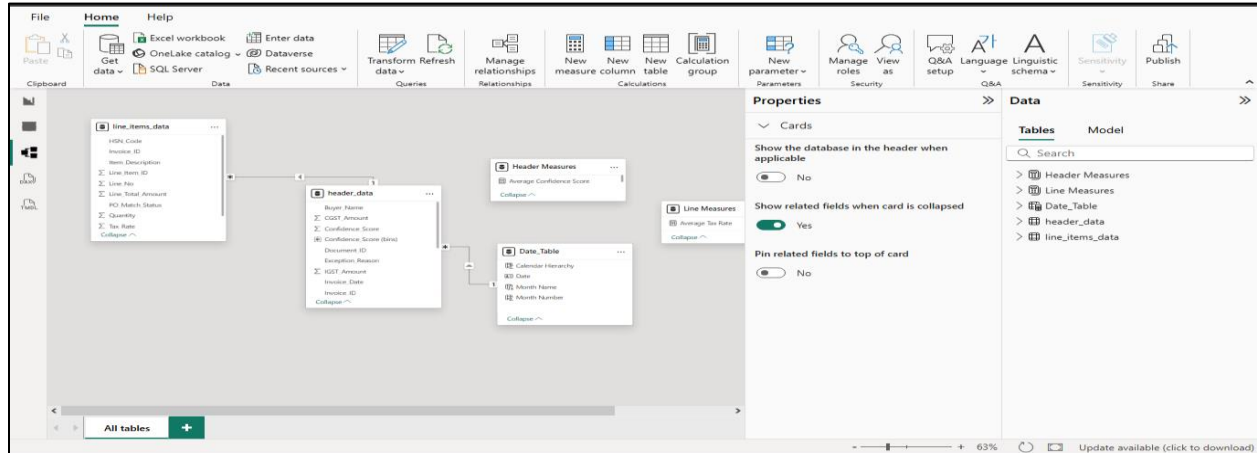
The Power BI model consists of:

- Header_Data
- Line_Items_Data
- Date_Table
- Header Measures Table
- Line Items Measures Table

Relationships:

- Header_Data → Line_Items_Data
- Date_Table → Header_Data

Figure 8.1: Model View: Relationship Tables Sample



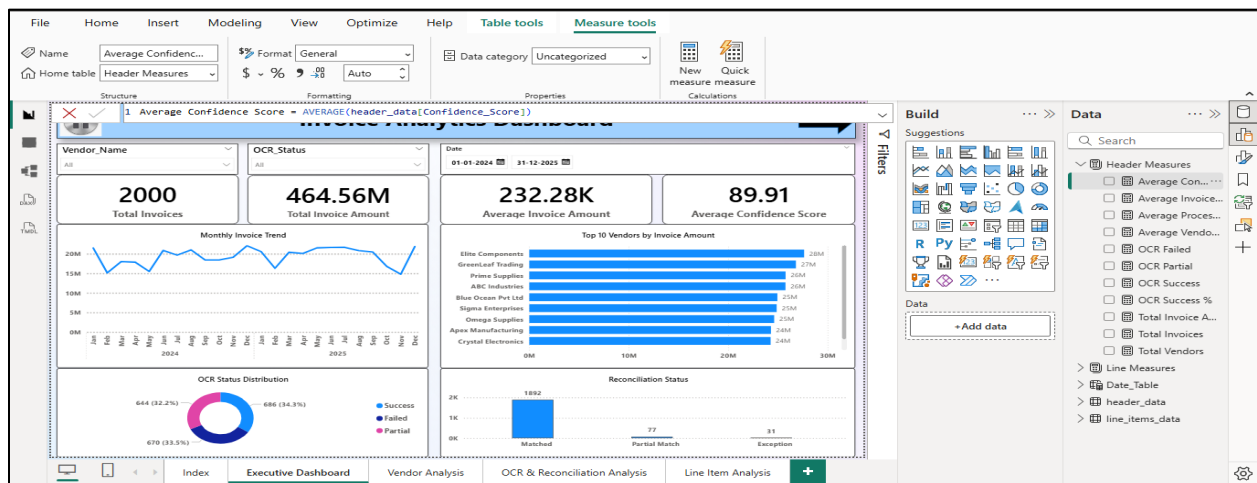
8.3: DAX Measures

DAX measures were created to calculate business KPIs.

Examples:

- Total Invoices
- Total Invoice Amount
- Average Invoice Amount
- Average Confidence Score
- Total Vendors
- OCR Success %

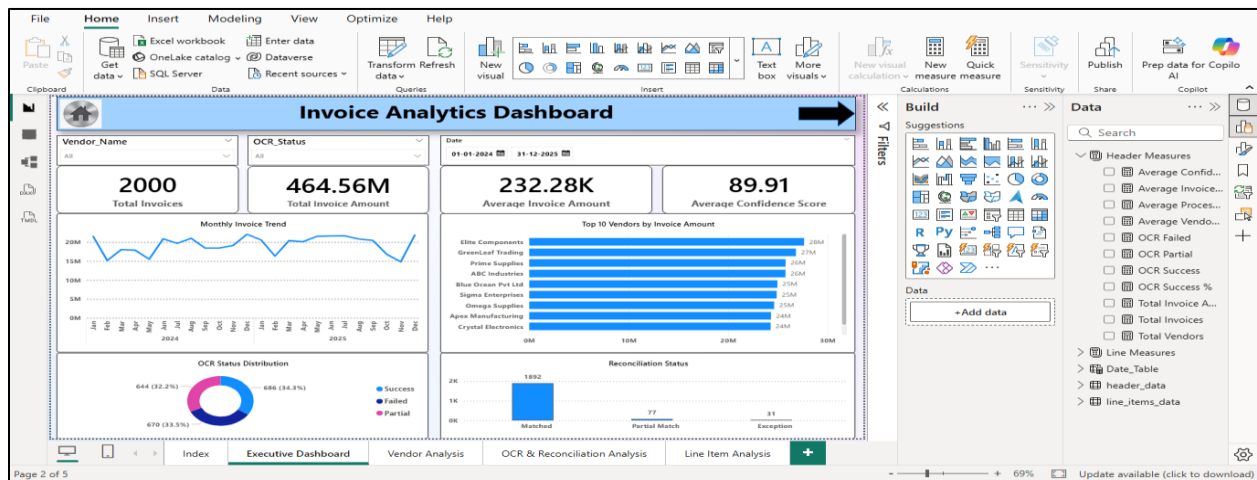
Figure 8.2: DAX formula in measure table Sample



8.4: Dashboard Pages

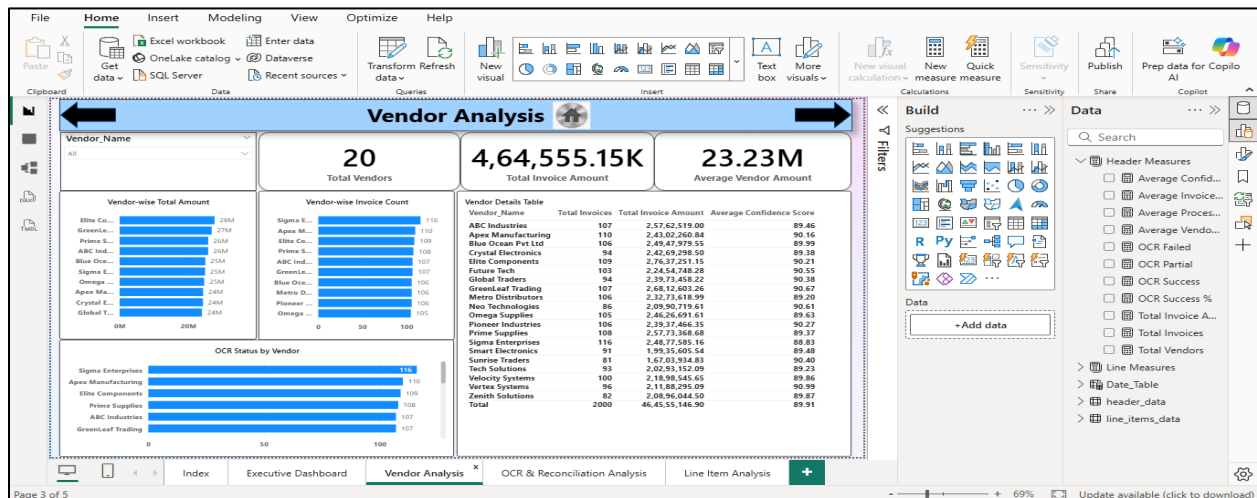
Executive Dashboard: Provides an overall summary using KPI cards, trend analysis, top vendors, OCR status, and reconciliation status.

Figure 8.3: Executive Dashboard Sample



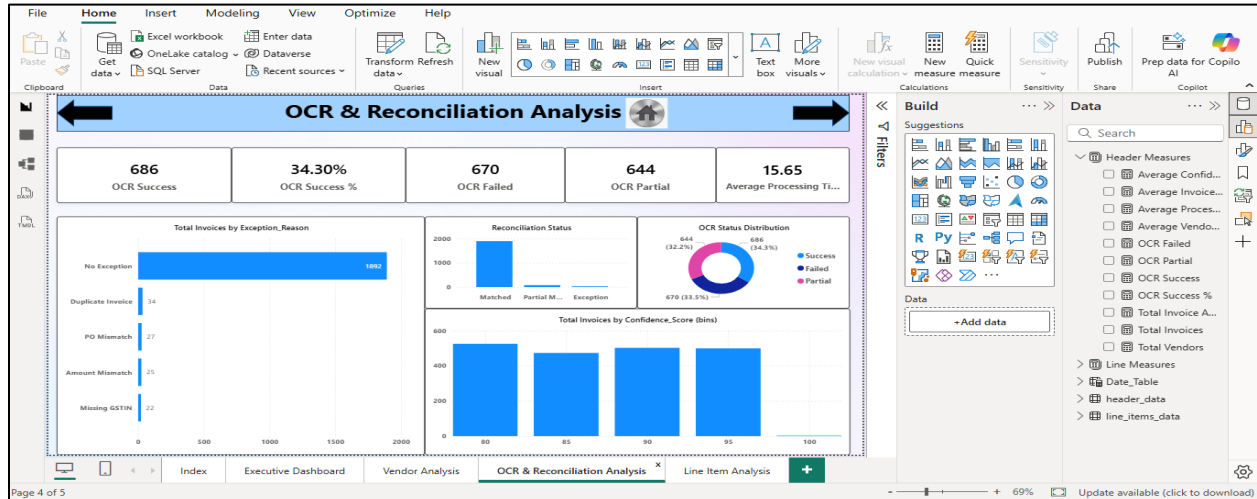
Vendor Analysis: Analyzes vendor performance using invoice count, invoice amount, confidence score, and vendor details.

Figure 8.4: Vendor Analysis Dashboard Sample



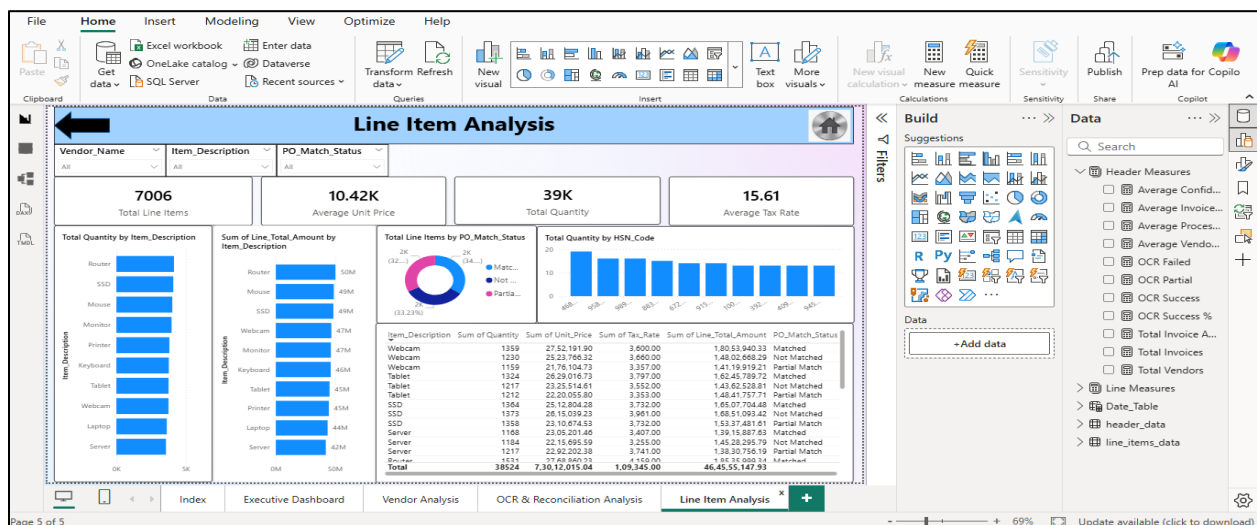
OCR & Reconciliation Analysis: Monitors OCR performance, reconciliation status, confidence score distribution, and exception reasons.

Figure 8.5: OCR & Reconciliation Dashboard Sample



Line Item Analysis: Analyzes purchased items, quantities, PO match status, HSN codes, and line-item details.

Figure 8.6: Line Item Analysis Dashboard Sample



Analysis 9: Business Insights

Key insights obtained from the analysis include:

- Invoice processing trends varied across months.
- A small number of vendors contributed a large share of the invoice value.
- Most invoices were processed successfully through OCR.
- Confidence scores were generally high, indicating reliable OCR extraction.
- Most invoices were successfully reconciled, with only a few exceptions requiring manual review.
- Line-item analysis identified the highest-value items and the most frequently purchased products.

Analysis 10: Challenges & Solutions

Challenge	Solution
CSV Import Issues	Corrected encoding and date formats
Data type mismatches	Converted columns to appropriate data types
SQL relationships	Implemented Primary and Foreign Keys
Python–MySQL connection	Used PyMySQL and SQLAlchemy
Power BI date analysis	Created a dedicated Date Table

Analysis 11: Conclusion

This project successfully demonstrated a complete end-to-end data analytics workflow using Excel, MySQL, Python, and Power BI. The solution covered data preparation, database design, SQL analysis, Python-based data processing, and interactive dashboard development. The project strengthened practical skills in data analysis, business intelligence, and reporting while generating meaningful business insights from invoice data.

Analysis 12: Future Enhancements

Possible future improvements include:

- Integrating live database connections for real-time reporting.
- Automating ETL using Python.
- Publishing Power BI reports to the Power BI Service.
- Implementing Row-Level Security (RLS).
- Adding forecasting and predictive analytics using machine learning.
- Scheduling automatic data refresh.